# Arms: A Router for Multilingual Vision-Language Model Selection

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04 Apr 2025 (modified: 05 Jul 2025) Submitted to ACMMM 2025
Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors Revisions (/revisions?id=Rl9sF97EQO) BibTeX
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**Abstract:**
Recent advances have demonstrated the potential of model selection strategies that dynamically choose the most suitable paid language model (LM) for a given input, balancing predictive performance and computational cost. However, these methods are typically limited to unimodal textual inputs and fail to extend to the growing number of multimodal applications involving both images and text. In this work, we address this gap by introducing a large-scale, multilingual benchmark for vision-language model (VLM) selection, which we refer to as \textbf{M$^3$}. This dataset comprises 56,107 image-text pairs spanning 8 languages, annotated with outputs from four open-source and three commercial VLMs across four representative benchmarks. Building upon this resource, we propose \textbf{Arms} (\textbf{a} \textbf{r}outer for vision-language \textbf{m}odel \textbf{s}election), a mixture-of-experts based routing framework that adaptively selects the most suitable VLM for each input by leveraging attention-based multimodal fusion mechanisms. We conduct a systematic analysis on how the size of training data influences router performance and introduce two scalable strategies—incremental training and independent training—to improve Arms' generalization to previously unseen VLMs. Extensive experiments on both in-distribution and out-of-distribution test sets show that Arms consistently outperforms individual VLMs and even surpasses GPT-4o when the selection space is expanded using our proposed strategies. All code, models, and datasets will be made publicly available to facilitate future research.

**Keywords:** Vision-Language Models, Model Selection, Multilingual Dataset, Router Training
**Primary Subject Area:** [Content] Vision and Language
**Secondary Subject Area:** [Generation] Generative Multimedia
**Submission Guide:** Yes
**Open Discussion:** Yes
**Author Registration Confirmation:** Yes
**Reviewer Participation:** Yes
**Submission Number:** 5622

**Discussion (?id=Rl9sF97EQO#discussion)**

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## Paper Decision

Decision by Program Chairs 05 Jul 2025, 04:14 (modified: 05 Jul 2025, 06:00) Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Authors

Revisions (/revisions?id=AycOGFDQJM)

**Decision:** Reject

## Meta Review of Submission5622 by Area Chair Sr51

Meta Review by Area Chair Sr51 26 Jun 2025, 14:50 (modified: 05 Jul 2025, 05:17)

Senior Area Chairs, Area Chairs, Authors, Reviewers, Program Chairs Revisions (/revisions?id=akPr4oegjw)

**Metareview:**

This paper introduces Arms, a router for multilingual vision-language model (VLM) selection, along with M^3, a new large-scale multilingual benchmark dataset comprising 56,107 image-text pairs in 8 languages. The authors propose a mixture-of-experts (MoE)-based framework that adaptively selects optimal VLMs using attention-based multimodal fusion.

Reviewers provided mixed evaluations (ratings ranged from strong rejection [2] to marginally above acceptance [6]). Strengths identified include the novelty of proposing a multilingual benchmark and the empirical improvements achieved by the MoE-based routing mechanism. However, major concerns consistently emerged around the limited methodological novelty (the MoE architecture combining known components such as BERT and CLIP), insufficient consideration of practical factors such as computational cost and inference latency, incomplete multilingual analysis (especially regarding low-resource languages), and inadequate baseline comparisons and ablation studies.

Authors did not provide rebuttal. The AC reviewed the manuscript and reviewer comments, agreeing with the substantial criticisms raised by reviewers. Consequently, the AC recommends rejection.

**Recommendation:** Reject

**Confidence:** 5: The area chair is absolutely certain

## Arms: A Router for Multilingual Vision-Language Model Selection

Official Review by Reviewer ydDS 25 May 2025, 19:36 (modified: 05 Jul 2025, 05:22)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Authors Revisions (/revisions?id=dBtaPhC6u5)

**Review:**

1. Strengths:

(1). First large-scale benchmark for VLM selection across multiple languages.

(2). MoE design with attention-based fusion delivers clear accuracy gains.

2. Weakness:

(1). The router's goal is reduced to predicting binary correctness, omitting cost and latency considerations that motivated model selection. The author should report the incorporation of estimated inference cost or response time into routing decisions.

(2). The paper reports aggregate metrics but provides no examples of queries that are misrouted or where the router underperforms individual VLMs, obscuring understanding of its limitations.

(3). Although eight languages are included, half of them each account for less than 3% of M3. The router's per-language performance, especially on truly low-resource languages, is not separately evaluated.

**Rating:** 4: Ok but not good enough - rejection
**Confidence:** 4: The reviewer is confident but not absolutely certain that the evaluation is correct
**Fit:** Relevant to a part of the community
**Fit Justification:**
None

**Technical Quality:** 5
**Presentation Quality:** 4
**Rebuttal Questions:**
See the Weakness

**Additional Comments:**
None

## Review for Paper #5622

Official Review by Reviewer 724Z 20 May 2025, 00:06 (modified: 05 Jul 2025, 05:22)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Authors Revisions (/revisions?id=z9kkaklItY)

**Review:**

### Summary

This paper proposes Arms, a routing framework for multilingual and multimodal Vision-Language Model (VLM) selection.

### Pros

1. Novel task: Extend routing from text-only to multilingual, multimodal VLM selection.
2. Large-scale dataset: Introduces $M^3$, the largest VLM selection dataset to date.
3. Strong results: Arms consistently outperforms all individual models under both in-distribution and out-of-distribution settings.

### Cons

1. Limited methodological novelty: While the task is novel, the architecture (BERT + CLIP + MoE + attention) is a straightforward combination of known components.
2. Missing cost/latency consideration: Although the authors mention hardware variability, in practice, latency and inference cost are crucial for model selection and should be factored in for real-world applicability.
3. Weak baselines: The paper does not adapt existing LLM selection methods (e.g., FORC, LLM-Blender) for multimodal settings as comparative baselines. Comparing only against individual VLMs limits the evaluation depth.

**Rating:** 6: Marginally above acceptance threshold

**Confidence:** 4: The reviewer is confident but not absolutely certain that the evaluation is correct
**Fit:** Relevant to a part of the community
**Fit Justification:**
The paper addresses multimodal model selection involving both vision and language, introduces a large-scale multilingual dataset, and proposes a routing framework for VLMs—all of which are core themes in ACM Multimedia, including multimodal systems, multilingual media processing, and intelligent model adaptation.

**Technical Quality:** 7
**Presentation Quality:** 7
**Rebuttal Questions:**

1. The router is trained using binary labels (whether a VLM answers correctly), which may overlook nuances in answer quality. How did you ensure label quality in the $M^3$ dataset?
2. English accounts for 46.05% of the data, much higher than other languages. Have you evaluated the impact of this imbalance on multilingual performance? How does the router perform on low-resource languages like Serbian (1.47%)?
3. Incremental training performs better with smaller initial datasets （Figure 7(a)）. Can you provide more analysis on why this happens?
4. The number of MoE experts is fixed (4 or 7). Can Arms scale to support dozens of VLMs? Have you conducted ablation studies on the expert count and its effect on performance and efficiency?
5. What is the computational overhead of Arms itself? Compared to directly using a single VLM, is the routing cost justified by the performance gain?
6. How does Arms handle queries that none of the VLMs can answer correctly? Is there a fallback or rejection mechanism for low-confidence predictions?

**Additional Comments:**
It would be helpful to provide a more detailed breakdown of inference time and system overhead to better assess the practical feasibility of deploying Arms in real-world applications.

## multilingual benchmark for vision-language model

Official Review by Reviewer dVZt 17 May 2025, 19:19 (modified: 05 Jul 2025, 05:22)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Authors Revisions (/revisions?id=vtcjtCoxbl)

**Review:**
The paper proposes a multilingual benchmark for vision-language model (VLM) selection. This dataset comprises 56,107 image-text pairs spanning 8 languages, annotated with outputs from four open-source and three commercial VLMs across four benchmarks. It also propose a mixture-of-experts based routing framework for VLM selects. Incremental training and independent training are proposed. Experimental results show the framework consistently outperforms individual VLMs.

It is unfortunately that there are no multilingual related studies and experiments. The research questions raised in the paper can be explored in a non-multilingual benchmark, which weakens the value of the proposed benchmark. The comparison in the paper is unfair, since the proposed model is trained by some samples, it means the test on your model is not zero-shot, whilr other single VLMs are zero-shot.

**Rating:** 5: Marginally below acceptance threshold
**Confidence:** 4: The reviewer is confident but not absolutely certain that the evaluation is correct
**Fit:** Very clearly a multimedia paper, of interest to a large part of the community
**Fit Justification:**
The proposed benchmark is not well explored on its multilingual characteristics. The experiments are not convincing.

**Technical Quality:** 5

**Presentation Quality:** 5
**Rebuttal Questions:**

1. Studies specially on multilingual are expected, since the proposed benchmark is a multilingual one. Why there are no multilingual related studies and experiments.
2. How to choose threshold $\theta$ ?
3. Abalation study on the MoE module is expected, as well as the name.

**Additional Comments:**
Both the query and the expert use same label $i$ in equation (2) (3), which is confusing. It is better to refer to corresponding figure or table in first paragraph ( 5.2.1, 5.3.1 and so on) .

## review for Arms

Official Review by Reviewer QYFo 16 May 2025, 14:05 (modified: 05 Jul 2025, 05:22)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Authors Revisions (/revisions?id=oBEgSbebnF)

**Review:**
This paper presents a contribution to the field of vision-language model (VLM) selection. It addresses a gap in existing research by introducing a large-scale multilingual benchmark (M^3) and proposing Arms, a routing framework for VLM selection.

Strengths:

1. It introduces a novel multilingual benchmark.
2. It presents a routing framework for VLM selection.

Weaknesses:

1. The literature review is incomplete. Many works on multilingual multimodal models, such as VisCPM[1], PaLI[2], and Ziya-Visual[3], are not discussed.
2. The proposed method lacks novelty. There are already numerous similar approaches for model selection, like [4-6].
3. Many crucial ablation experiments are absent. For example, there is no exploration on how to determine the number of experts.
4. The paper dose not explain the rationale behind the selection of the current models. From my perspective, excellent multimodal models such as InternVL[7], LLaVA-Next[8], Ovis[9], and Ola[10] should also be included.
5. Although the authors propose a new benchmark, it is necessary to conduct evaluations on previous model selection benchmarks to verify the effectiveness of the proposed method.
6. The paper does not address the computational cost of using Arms, particularly when more VLMs are added to the selection space.
7. The authors assert that Arms can generalize to unseen VLMs, yet the experiments mainly concentrate on a limited number of VLMs. Testing Arms on a more diverse range of models would be beneficial to comprehensively assess its generalization ability.

[1] Large Multilingual Models Pivot Zero-Shot Multimodal Learning across Languages. [2] PaLI: A Jointly-Scaled Multilingual Language-Image Model. [3] Ziya-Visual: Bilingual Large Vision-Language Model via Multi-Task Instruction Tuning. [4] Mixture of LoRA Experts. [5] Smoothie: Label - Free Language Model Routing. [6] PolyRouter: A Multi - LLM Querying System. [7] InternVL3: Exploring Advanced Training and Test-Time Recipes for Open-Source Multimodal Models. [8] LLaVA-NeXT: Open Large Multimodal Models. [9] Ovis: Structural Embedding Alignment for Multimodal Large Language Model. [10] Ola: Pushing the Frontiers of Omni-Modal Language Model with Progressive Modality Alignment.

**Rating:** 2: Strong rejection
**Confidence:** 4: The reviewer is confident but not absolutely certain that the evaluation is correct
**Fit:** Relevant to a part of the community

**Fit Justification:**
The paper deals with VLMs, which are relavant to multimedia research. The construction of a multilingual multimodal dataset and the development of a router for selecting such models are relevant to the multimedia research community to some extent.

**Technical Quality:** 6
**Presentation Quality:** 6
**Rebuttal Questions:**
See Weaknesses.

**Additional Comments:**
I hope the authors further explore the optimization of the Arms router's performance and delve deeper into the computational cost analysis. Additionally, expanding the evaluation to a more diverse set of vision - language models would strengthen the claim of the router's generalization ability.

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